

Nicolas Bonasoro

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Aerospace Simulation | Space Robotics | Lunar Surface Systems | Terrain Interaction & Verification

EDUCATION

Embry-Riddle Aeronautical University

Bachelor of Science in Aerospace Engineering (Dean's List)

Daytona Beach, FL

December 2026

Rockland Community College

Associate of Applied Science in Business Administration

Suffern, NY

May 2021

COURSE PROJECT

South Pole Lunar Relay Satellite and Ground Station | *Senior Design, ERAU*

Jan 2026 – Present

- Led systems engineering for a lunar South Pole relay satellite and ground station to support lunar surface missions
- Developed S-band/X-band link budget analysis and supported orbital analysis, ADCS design, CATIA satellite bus design, thermal testing, launch vehicle selection, and Interface Control Document (ICD) development

RESEARCH EXPERIENCE

Robust Autonomous Lunar Landing | *Space Robotics & Generative Estimation Lab, ERAU*

Oct 2025 – Present

- Developed Chrono::DEM and Chrono::CRM lander simulations to evaluate touchdown behavior, sinkage, slippage, and tip-over risk
- Measured higher leg penetration in settled regolith (0.53 m) than unsettled regolith (0.34 m) and analyzed slope/yaw cases for SE(3) dynamics and machine-learning training

Lunar Trafficability NASA FSGC | *Space Robotics & Generative Estimation Lab, ERAU*

Sep 2025 – Present

- Led Chrono::DEM and Chrono::CRM cone penetration testing simulations to evaluate lunar regolith force-depth behavior
- Coordinated lunar trafficability research workflow through Jira task tracking, researcher mentoring, simulation data analysis, and support for rover-mounted payload assembly

Mars & Moon Landing Site Estimation | *Space Robotics & Generative Estimation Lab, ERAU*

Feb 2025 – Apr 2025

- Developed Python workflows to extract 3D Mars terrain meshes from NASA MRO CTX DEM data and generate virtual environments for rover simulation
- Simulated autonomous rover missions with ROS 2 and NVIDIA Omniverse to evaluate path-planning behavior and landing-site hazard constraints

ACADEMIC PAPERS & PRESENTATIONS

- **N. Bonasoro**, J. Cortese, and C. Kilic, "Terramechanics-Based Comparison of DEM and CRM for Lunar Touchdown and Cone Penetration Applications," extended abstract under review, AIAA SciTech (2027)
- J. Cortese, **N. Bonasoro**, J. Graff, D. Zhang, M. Salih, G. Ellithy, and C. Kilic, "Closing the Loop on In-Situ Lunar Trafficability," extended abstract under review, AIAA SciTech (2027)
- R. Nagarajan, E. Girgin, **N. Bonasoro**, J. Nakaldov, C. Kilic, and D. Canales Garcia, "AI-Enabled Physics-Based Framework for Tip-Over Risk Prediction in Lunar Lander Touchdown," paper submission and oral presentation accepted for the 77th International Astronautical Congress (2026)
- R. Nagarajan, E. Girgin, D. Canales Garcia, J. Nakaldov, **N. Bonasoro**, and C. Kilic, "Geometric Modeling of Multi-Contact Touchdown Dynamics and Stability of Lunar Landers," paper submission and oral presentation accepted for the 77th International Astronautical Congress (2026)
- **N. Bonasoro**, J. Cortese, and C. Kilic, "Terramechanics-Based Comparison of DEM and CRM for Lunar Surface Application," Poster Presentation, Embry-Riddle Aeronautical University's Discovery Day (2026)

SKILLS

Languages: C++, CUDA, Python, MATLAB, LaTeX

Tools: Linux (Ubuntu), Docker, CMake, Git, GitHub, Shell, Visual Studio, ParaView, Blender, Simulink, Unreal Engine, MS Office, Jira, Overleaf

Simulation: CATIA V5, ROS 2, NVIDIA Omniverse, Femap, ANSYS Fluent, Project Chrono (Chrono::DEM (Discrete Element Method), Chrono::CRM (SPH-based continuum deformable terrain model))